

Nathaniel Sawant

Available Jan 2025 – Aug 2025

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EDUCATION

Northeastern University, Khoury College of Computer Sciences

Boston, MA

Bachelor of Science in Computer Science, Concentration in Artificial Intelligence

Sep. 2021 – Dec. 2024

GPA: 3.71/4.00

Leadership: Northeastern U. IEEE Chapter (President, *Fall 2023*): NUHOC (VP Publicity, *Summer 2024*)

Relevant Courses: Software Engineering, Database Management Systems, Machine Learning 1 & 2, Object-Oriented Design, Algorithms & Data Structures, C++ Programming, Natural Language Processing, Artificial Intelligence, Computer Systems, Mathematics of Data Models, Foundations of Data Science, Probability & Statistics

Clubs: IEEE (Professional), NUHOC (Outdoors), NUPiC (Photography), Generate (Product Development)

PROFESSIONAL EXPERIENCE

Software Engineer

Jan. 2024 – Jun. 2024

Babel Street, Unstructured Team

Somerville, MA

- Doubled download speeds with **multi-threading** & cut memory by 20% in Named Entity Recognition training
- Analyzed NER categorical changes of 2M+ items with Python & ontology, preventing malformed/incorrect data
- Dockerized Maven projects and created Jenkins CI/CD pipeline to perform comprehensively test dependencies

Software Data Engineer

Jan. 2023 – Jun. 2023

Fidelity Investments, Advanced Data Analytics Team

Boston, MA

- Streamlined development with a **shared code snippet tool** for VSCode/Jupyter Lab using **TypeScript**
- Conducted **ETL processes** on 10+ sites to **scrape data points** to SQL for over 65,000 items with **Python**
- Created **data visualization dashboard** to track usage of **100+ datasets**, enabling informed decision-making

Research Assistant

Oct. 2021 – Aug. 2022

Northeastern University, Rehabilitation Games & Extended Reality Lab

Boston, MA

- Developed two player gameplay mechanics using **Unity 3D engine**, OpenXR, & Photon Unity Networking
- Captured participants' **eye-tracking data at 60HZ** with Vive Pro to analyze where locations of focus

EXTRACURRICULAR EXPERIENCE

Project Lead (DevDB, Snapper)

May. 2024 – Present

Generate Product Development at Northeastern University

Boston, MA

- Led and collaborated with 5-13 person Agile team to manage **project priorities, deadlines, & deliverables**
- DevDB: Defined system design** for a development database platform using **Node.js, AWS, & PostgreSQL**
- Snapper:** Analyzed 5 competitors and coordinated with client, design, tech leads to plan & develop 6 deliverables

Software Engineer (Voxeti, Tubender)

Sep. 2023 – Apr. 2024

Generate Product Development at Northeastern University

Boston, MA

- Voxeti:** Designed **RESTful APIs** using Golang & MongoDB for user-run 3D print-on-demand website
- Tubender:** Implemented onboarding flow and settings page frontend using **React & Electron** with C++ backend

PROJECTS

Fish Species Computer Vision Classifier | *Python, PyTorch, FastAI*

Sep. 2024 – Present

- Building multi-class classification models capable of predicting correct species from 199 fish species
- Implementing deep learning pipelines for **Convolutional Neural Network and Visual Transformers**

Deep Reinforcement Learning Agent | *PyTorch, Unity, C#*

Mar. 2023

- Trained and tested **reinforcement learning agent** that seeks target in unknown continuous environment
- Built procedurally generated maze with **Unity for training and testing** of model over hundreds of epochs

TECHNICAL SKILLS

Languages: Python, Java, C#, Golang, C/C++, SQL, Java/TypeScript, HTML/CSS

Frameworks: React, Next.js, Unity, JUnit, FastAPI, Express

Developer Tools: Git, Docker, AWS (S3, EC2, RDS), NoSQL (MongoDB), Relational Databases (Oracle, Postgres)

Libraries: pandas, NumPy, Matplotlib, scikit-learn